COURSE WORKBOOK

Building Web Apps with ArcGIS® Experience Builder

Official Esri Training Courseware

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Building Web Apps with ArcGIS® Experience Builder

STUDENT EDITION

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Course introduction

ArcGIS Experience builder is Esri's next-generation app builder that enables you to create, customize, and share tailored mapcentric or datacentric apps for computer, tablets, or mobile devices. Apps created with ArcGIS Experience Builder can be dynamic and update with user interactions to further meet your audience's needs.

In this course, you will learn how to create both mapcentric and datacentric apps in ArcGIS Experience Builder. You will configure widgets and actions to customize the app for your needs as well as adjust the app layout. You will also discover how to adapt apps for multiple device types and how to publish and share your apps for wider use.

Course goals

After completing this course, you will be able to perform the following tasks:

- Design the app layout and theme based on the audience and purpose.
- Configure widgets to enable users to interact with your organization's web maps and 2D and 3D data.
- Configure widgets to provide data-driven functionality across multiple pages.
- Test, preview, and publish your apps for use on a variety of devices.

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Some exercises in this workbook require data. Depending on the course format, the data is available on a DVD in the back of a printed workbook or as a data download. To use the data, extract it to your C:\EsriTraining folder.

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Icons used in this workbook

Estimated times provide guidance on approximately how many minutes an exercise will take to complete.

Notes indicate additional information, exceptions, or special circumstances about specific course topics.

Recommended practices improve efficiency and save time.

Esri Academy resources provide more in-depth training on related topics.

Additional resources provide additional information about related topics.

Warnings indicate potential problems or actions that should be avoided.

1 Introducing ArcGIS Experience Builder

In this lesson, you will discover the key capabilities of ArcGIS Experience Builder and observe these capabilities by exploring the functionality of an existing web app. You will also gain an understanding of how these capabilities compare to other ArcGIS app builders, such as ArcGIS Web AppBuilder. Additionally, this lesson will present the workflow for creating a web app with ArcGIS Experience Builder, which will help guide the content in this course.

Topics covered

- Web app capabilities App
- creation workflow

Capabilities of ArcGIS Experience Builder

With ArcGIS Experience Builder, data can be quickly transformed into compelling web apps without writing a single line of code. ArcGIS Experience Builder provides the following capabilities for building a web app.

- **Interactive, data-driven content:** Incorporate interactive content that is powered by existing 2D and 3D data.
- **Flexible layout options:** Use the flexible layout options to create purpose-driven web apps. Web apps can be created as a single web page or a multipage website. Each page can contain mapcentric content, datacentric content, and static content, such as text and images.
- **Modern drag-and-drop interface:** Take advantage of a modern drag-and-drop interface, rather than code, to customize the appearance of a web app.
- **Internet best practices:** Apply current internet best practices by creating web apps that are built with ArcGIS API for JavaScript 4.x and incorporate the latest web technology.
- **Mobile-friendly:** Deploy mobile-friendly apps that are responsive and functional on any device.
- **New functionality:** Integrate the latest ArcGIS functionality, such as maps created with ArcGIS Map Viewer.

Capabilities of ArcGIS Experience Builder (continued)

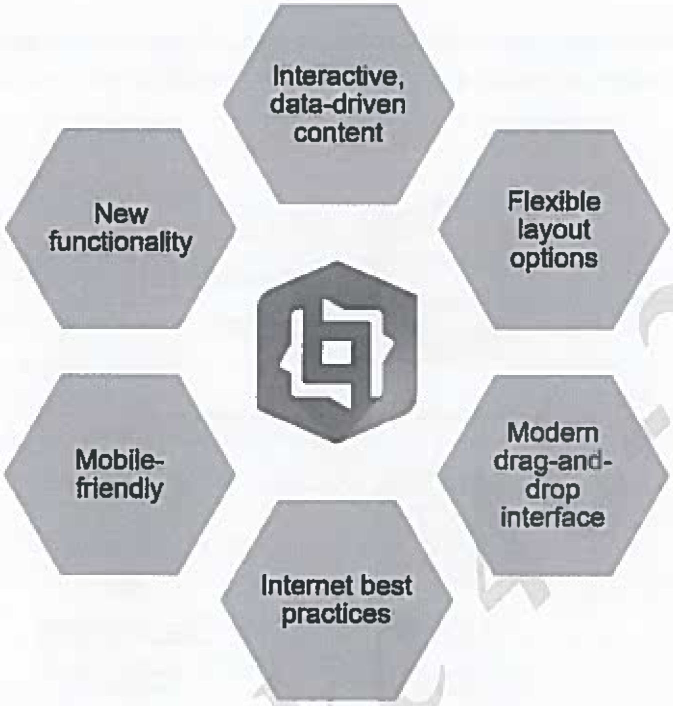

Figure 1.1. There are many web app capabilities that are supported by ArcGIS Experience Builder.

Introduction to ArcGIS Experience Builder versions

To take advantage of the ArcGIS Experience Builder capabilities, your organization must choose a version to implement. There are three versions of ArcGIS Experience Builder: ArcGIS Online, ArcGIS Enterprise, and developer edition.

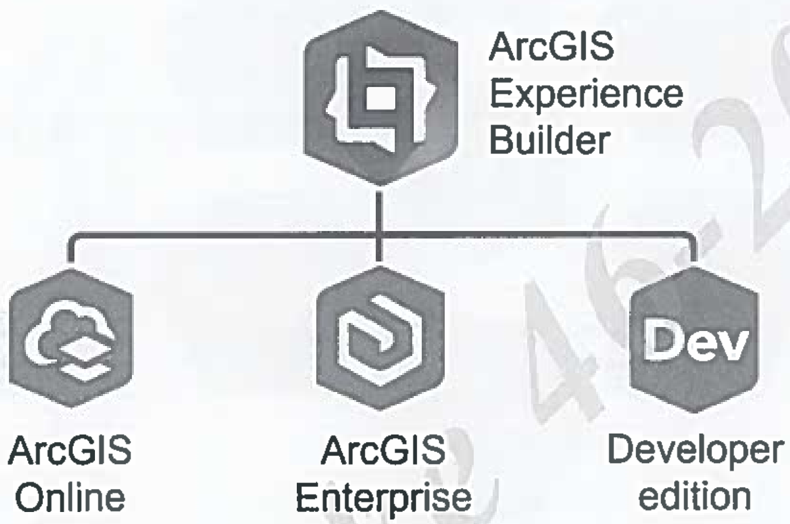

Figure 1.2. There are three versions of ArcGIS Experience Builder that can be used: ArcGIS Online, ArcGIS Enterprise, and developer edition.

Version descriptions

In most cases, your organization will choose the version that aligns with the type of portal that is being used, such as ArcGIS Online or ArcGIS Enterprise. Your organization may consider the developer edition if additional or advanced functionality is required. The following table describes each version of ArcGIS Experience Builder.

Introduction to ArcGIS Experience Builder versions (continued)

ArcGIS Experience Builder version	Description
ArcGIS Online	• Included with ArcGIS Online • Updated automatically as part of the ArcGIS Online release cycle • Used in this course
ArcGIS Enterprise	• Included with ArcGIS Enterprise version 10.8.1 and later (ArcGIS Enterprise is similar to ArcGIS Online); however, this highly customizable product is fully controlled and managed by the organization • Updated when ArcGIS Enterprise is upgraded to a new version
Developer edition	• Supported in ArcGIS Online and ArcGIS Enterprise 10.6 and later • Updated when a newer version is installed • Advanced functionality that enables the creation of custom widgets, themes, and actions*

*This advanced functionality requires a connection to ArcGIS Online or ArcGIS Enterprise version 11.0 and later.

ArcGIS Experience Builder Help: *About release versions*

Explore an ArcGIS Experience Builder app

Exploring the functionality of an example web app is the best way to learn about the capabilities of ArcGIS Experience Builder. You will visit the ArcGIS Experience Builder Gallery and familiarize yourself with an example web app to gain ideas for using ArcGIS Experience Builder capabilities in your own work.

Instructions

a. In a web browser, go to the ArcGIS Experience Builder Gallery. In the

b. Gallery, review the different web apps that are available. Click a web

c. app that interests you.

> If the link takes you to a case study page, click the link for the web app on the page or choose a different app.

d. Answer the following questions using the selected web app.

e. When you are finished, close your web browser.

1. **What is the name of the web app that you chose?**

2. **What functionality is provided by the web app?**

3. **If applicable, in what ways do the elements in the web app interact with data?**

4. **Having explored a web app, how could you use an ArcGIS Experience Builder web app in your own work?**

Use cases for ArcGIS Experience Builder

Many of you are likely aware of other ArcGIS app builders that can be used to create a web app. While ArcGIS Experience Builder is the most powerful and flexible ArcGIS app builder, other options may also apply to a given use case depending on the required functionality.

App builder capabilities

The following table compares the capabilities of ArcGIS Experience Builder to the capabilities of ArcGIS Instant Apps and ArcGIS Web AppBuilder.

ArcGIS Experience Builder	ArcGIS Instant Apps	ArcGIS Web AppBuilder
Integrate data besides mapcentric content	**No** (Integrate mapcentric content only)	**No** (Integrate mapcentric content only)
Create multipage web apps	**No** (Create single-page apps only)	**No** (Create single-page apps only)
Design the layout of the content	**No** (Use the layout defined by the template)	**No** (Use the layout defined by the theme)
Use widgets to customize the app	**No** (Use preconfigured functionality only)	**Yes** (Use widgets to customize the app)
Incorporate ArcGIS Map Viewer maps	**Yes** (Incorporate ArcGIS Map Viewer maps)	**No** (Incorporate ArcGIS Map Viewer Classic maps only)
Use ArcGIS API for JavaScript 4.x	**Yes** (Use ArcGIS API for JavaScript 4.x)	**No** (Use ArcGIS API for JavaScript 3.x, which has limited long-term usability)

Use cases for ArcGIS Experience Builder (continued)

Use case considerations

ArcGIS Experience Builder is capable of creating web apps that fit a wide range of use cases. However, there are situations where a different ArcGIS app builder may be a better option.
Consider the following questions to determine whether a different app builder should be used.

1. **Is the required functionality compatible with an ArcGIS Instant App?** If there is an ArcGIS Instant App that aligns with the purpose of the web app without customization, you will likely want to use this app builder because it is the fastest and simplest option.
2. **Is ArcGIS Experience Builder missing any required functionality?** If so, you can build the app with ArcGIS Web AppBuilder. However, all versions of ArcGIS Web AppBuilder will be retired by the end of 2025, so all apps will need to be recreated with ArcGIS Experience Builder before this date. Otherwise, consider using Experience Builder developer edition to create widgets that provide the required functionality.

Esri Community Blog Post: *ArcGIS Web AppBuilder Roadmap for Retirement*

ArcNews: *How to Choose the Right Web App in ArcGIS Online*

ArcGIS Experience Builder creation workflow

Now that you understand the capabilities and usage of ArcGIS Experience Builder, you will discover the process for creating a web app. The workflow for creating a web app with ArcGIS Experience Builder consists of the following steps.

- **Plan** the web app by identifying the purpose and target audience for the app.
- **Design** the web app by outlining the design elements that will be provided on each page.
- **Configure** the web app by adding the design elements to a new experience in ArcGIS Experience Builder.
- **Customize** the web app by applying styles and adding actions.
- **Deploy** the web app by sharing the data and publishing the app.

The configure, customize, and deploy steps of the app creation workflow are iterative.

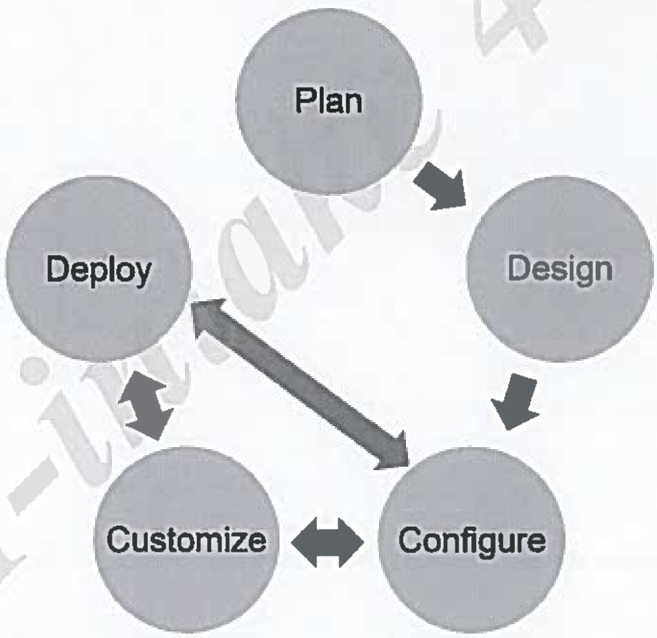

Figure 1.3. The workflow for creating a web app is a five-step process.

To better understand the workflow for creating a web app, the following sections visually compare an app design to a deployed app.

ArcGIS Experience Builder creation workflow (continued)

App design

This wireframe is an example design for a web app. The proposed layout of the widgets and the underlying data for the single-page app are shown on the wireframe. In this example, the web app will use a custom theme that will be applied during the configure step of the app creation workflow.

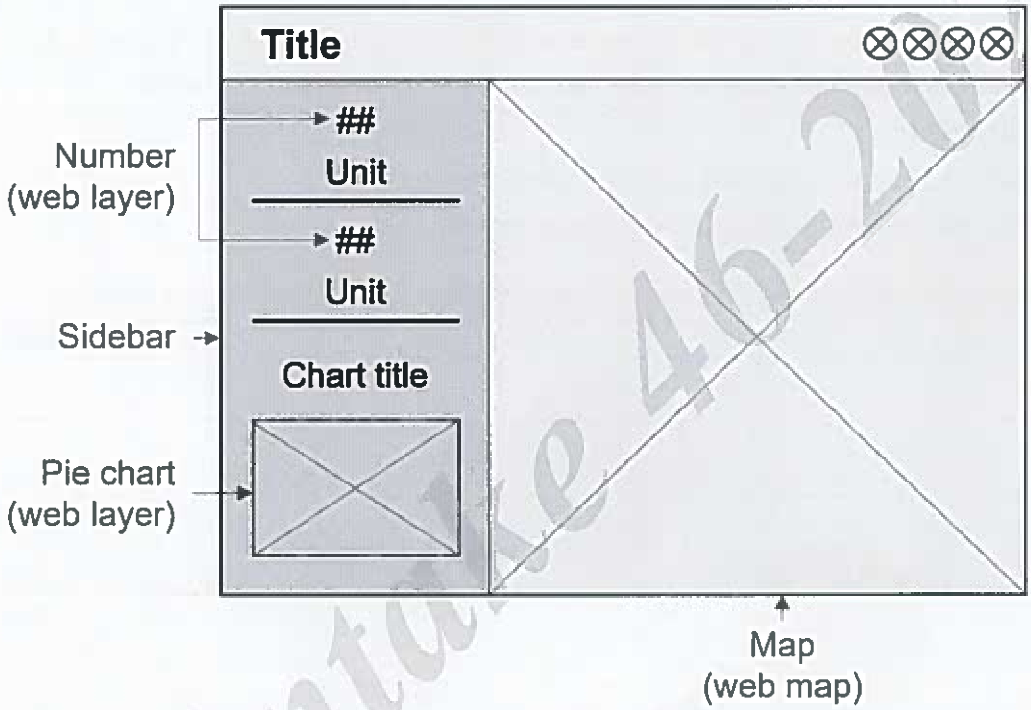

Figure 1.4. This wireframe diagram is an example design for a web app.

Deployed app

This example presents a deployed web app that was created based on the previously shown design.

ArcGIS Experience Builder creation workflow (continued)

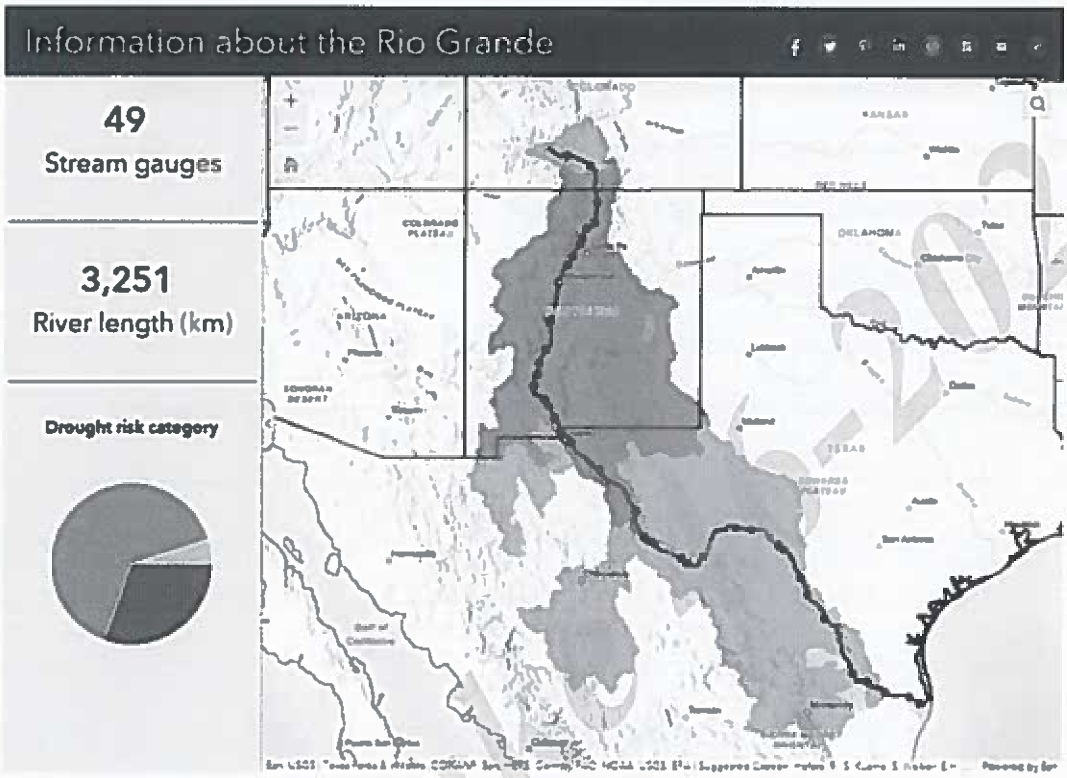

Figure 1.5. This example shows a deployed web app.

Lesson review

1. Which ArcGIS Experience Builder version enables you to build custom widgets?
   a. ArcGIS Experience Builder ArcGIS Online
   b. ArcGIS Experience Builder ArcGIS Enterprise
   c. ArcGIS Experience Builder developer edition
   d. ArcGIS Experience Builder JavaScript API edition
2. What capabilities distinguish ArcGIS Experience Builder from other app builders?

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Answers to lesson 1 questions

Explore an ArcGIS Experience Builder app (page 1-6)

1. What is the name of the web app that you chose?

   Answers will vary based on the chosen web app.

2. What functionality is provided by the web app?

   Answers will vary but may include the ability to show a splash screen, search a map or data, interact with custom buttons, or open multiple web pages.

3. If applicable, in what ways do the elements in the web app interact with data?

   Answers will vary but may include displaying data in a map, populating text based on the attributes of data layers, presenting data as a chart, or viewing a list of data layers.

4. Having explored a web app, how could you use an ArcGIS Experience Builder web app in your own work?

   Answers will vary based on personal experience. For example, you could create a web app that includes a map and elements that explain the data components.

2 Designing a web app

In this lesson, you will learn about the key elements of a web app design and the questions that you should consider when designing an app. You will also discover the process for choosing a starting template that aligns with the design for a planned web app. Finally, you will gain an understanding of how design principles are applied to an app by configuring a basic web app.

Topics covered

- Web app design
- Out-of-the-box templates
- Custom templates

Design elements

As part of the planning stage, the purpose and target audience for the web app should be specified. When designing a web app, you will need to identify the design elements that will help achieve the purpose of the app and thoughtfully arrange the elements to create a user-friendly experience for the target audience. The following list describes the key design elements of an ArcGIS Experience Builder web app.

- **Page layouts:** Sketch each page of the web app to show how the information will be organized. Alternatively, choose a template with a preconfigured layout that aligns with the purpose of the app and provides the required functionality. The layout of the page should consider the other design elements, such as the theme, widgets, and source content.
- **Theme:** Choose a theme that meets any branding requirements, promotes visual storytelling, and emphasizes the key information.
- **Widgets:** Identify the widgets that will be included on each page to provide the required functionality. Some widgets display static content, such as text and images, and other widgets provide interactive content.
- **Source content:** Select the source content, such as data and utility services, that will power the interactive widgets in the web app.

Design elements (continued)

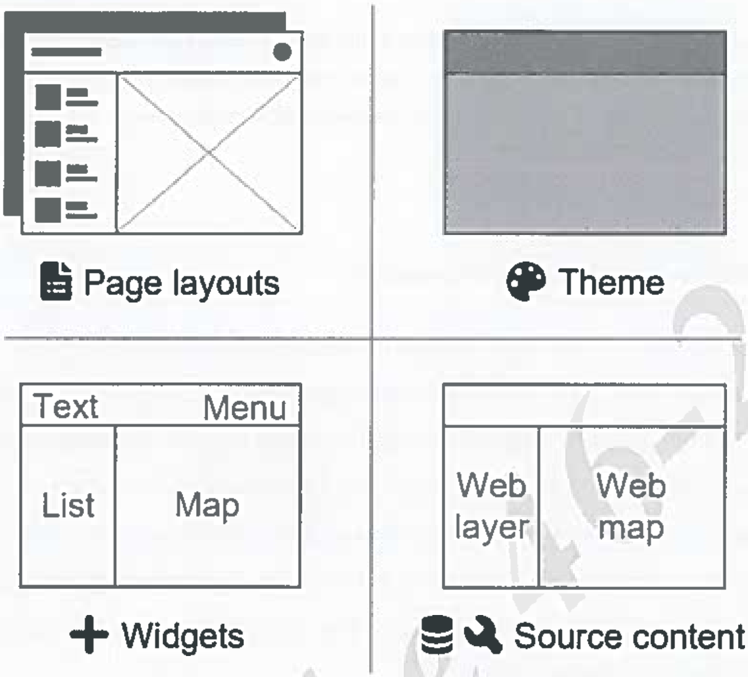

Figure 2.1. Page layouts, theme, widgets, and source content are the key design elements that affect the framework of a web app.

Design considerations

Previously, the four elements that should be included in a design for an ArcGIS Experience Builder app were identified. The four design elements are page layouts, theme, widgets, and source content. For each element, there are multiple questions to consider while developing the design.

Page layouts

- **Which template will be used to create the app?**

With ArcGIS Experience Builder, a starting template for the web app must be selected. The starting template will include at least one page, which will have a blank or preconfigured layout. The layout of the page will be sized to fill the device screen (full-screen app) or require scrolling to view all the content (scrolling page). When a web app is created from a starting template, pages can be added and removed. For blank pages in a web app, designing the layout from scratch can be time-consuming, but this approach provides the most opportunities for customization. For preconfigured pages in a web app, working with a preset layout can save time when designing the app, but this approach requires a firm understanding of the Enterprise Builder interface.

- **How will the target audience access the app?**

The types of devices that the target audience will use to access the app should be identified during the design phase to ensure that the layout supports a user-friendly experience.

Theme

- **What are the branding requirements for the app?**

If your organization requires the use of a standardized color scheme and font for all web apps, then a custom theme should be used rather than an out-of-the-box theme.

An administrator for the ArcGIS Online or ArcGIS Enterprise organization will need to configure the custom theme.

Otherwise, consider choosing a theme that visually conveys the purpose of the app. For

Design considerations (continued)

example, if the app displays information related to water resources, then a theme that incorporates blue colors may be a good choice.

- **What is the key information in the app?**

Consider using the theme colors and font to highlight the key information on the page. The theme color set contains eight color variables that can be used to categorize the information on the page: primary, secondary, light, dark, info, success, warning, and danger. For example, a hover effect can be configured for a button so that the fill changes to the info color variable when a user points to the button. Additionally, colored borders can be used to emphasize content on the page.

Widgets

- **What functionality does the app need to provide?**

The required functionality determines the types of widgets that should be included in the app.

Source content

- **Where will the source content come from?**

The configuration requirements for each widget determine the types of source content that will be needed, such as data and utility services. For each content source, identify the portal content that will be used and update the sharing level to make the content accessible to the target audience.

Exercise 2A

20 minutes

Explore the out-of-the-box templates

In this exercise, you will explore the available out-of-the-box starting templates in ArcGIS Experience Builder. Some starting templates include a page with a blank layout while others include one or more pages that have a preconfigured layout. Every starting template allows you to add and remove content, organize the content on the page, and customize the appearance of the web app.

In this exercise, you will perform the following tasks:

- View the available templates.
- Identify a template for a given scenario.

Step 1: Sign in to ArcGIS Experience Builder

You can access ArcGIS Experience Builder through the app launcher in ArcGIS Online or the landing page. For this exercise, you will use the landing page to sign in to the ArcGIS Online version of Experience Builder.

a. Open Firefox.

The web browser opens to the ArcGIS Experience Builder landing page.

b. Click Sign In.

c. In the ArcGIS Sign In dialog box, expand Your ArcGIS Organization's URL, if necessary. For the

d. organization's URL, type **trainingservices**, and then click Continue.

e. Click Your Course Account.

f. Type the organizational account username and password provided by your instructor, and then click Sign In.

g. Near the top-right corner of the page, click Create New.

When a new experience is created, you are prompted to select a starting template. The many template options are organized into the following tabs:

- **Default tab:** Shows the out-of-the-box templates that were created by the ArcGIS Experience Builder team
- **My Templates tab:** Shows the custom templates that are generated by you
- **My Favorites tab:** Shows the templates that you added to your list of favorites
- **My Groups tab:** Shows the custom templates that are shared with a group that you are a member of in ArcGIS Online
- **My Organization tab:** Shows the custom templates that are shared with your ArcGIS Online organization
- **ArcGIS Online tab:** Shows the custom templates that are shared with the public
- **Living Atlas tab:** Shows a curated selection of custom templates that are created by Esri and others

In this exercise, you will explore the available out-of-the-box templates on the Default tab.

Step 2: View the blank templates

First, you will view the out-of-the-box templates in the Blank category.

a. If necessary, click the Default tab to view the list of out-of-the-box templates. Next to

b. All, click Blank.

c. Point to each template and read the descriptions.

1. **When should a blank template be used?**

____

2. **Which template provides a fully blank layout that minimizes scrolling?**

____

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Step 3: Examine the Web AppBuilder Classic templates

Next, you will examine the out-of-the-box templates in the Web AppBuilder Classic category. These Web AppBuilder Classic templates closely align with the app themes that are available with ArcGIS Web AppBuilder. These templates help streamline the process for rebuilding classic web apps with ArcGIS Experience Builder.

a. Click Web AppBuilder Classic.

b. Point to each template and read the descriptions.

Now that you have reviewed the description for each template, you will choose the best Web AppBuilder Classic template for the following scenario.

***Scenario:** The Philadelphia Marathon Resources Web App was created using ArcGIS Web*

AppBuilder. You need to rebuild this web app using ArcGIS Experience Builder. The existing web app provides a simple interface with a header and two locations to access widgets through a controller widget.

You will learn more about widgets later in this course.

3. **Which ArcGIS Experience Builder template best aligns with the existing Philadelphia Marathon Resources Web App?**

___

There are slight differences between the layouts of the template and the existing web app. Unlike other ArcGIS app builders, the layout of a web app can be customized with ArcGIS Experience Builder, which is true regardless of the starting template that was selected.

Step 4: Explore the Map Centric templates

Next, you will explore the out-of-the-box templates in the Map Centric category, which includes templates from the Web AppBuilder Classic tab; ArcGIS Web AppBuilder is only capable of making web apps that contain mapcentric content.

a. Click Map Centric.

You will now discover the functionality of the Map Centric templates by choosing a template to use for three scenarios.

***Scenario 1:** You need to create a web app for city transportation engineers that has the ability to view a descriptive list of the proposed projects and navigate to the project locations in a map.*

b Review the descriptions for JewelryBox, Quick Navigation, and Pocket.

4. **For Scenario 1, which template should be used to create the planned web app?**

____

Hint: Choose a template that includes a list, not bookmarks.

***Scenario 2:** You need to create a web app for tourists that provides the ability to explore bookmarked locations of city landmarks, view charts that show peak travel times, and get directions to these locations.*

c Review the descriptions for Quick Navigation, Route, and Voyage.

5. **For Scenario 2, which template should be used to create the planned web app?**

____

Hint: Choose a template that includes bookmarks, charts, and directions.

***Scenario 3:** You need to create a web app for managers to track several metrics related to the condition of public works infrastructure based on routine inspections in the field.*

d Review the descriptions for Billboard, Data Collector, and Monitor.

6. **For Scenario 3, which template should be used to create the planned web app?**

____

Hint: Choose a template that includes key metrics and enables editing.

Step 5: Inspect the dashboard style templates

Next, you will observe the out-of-the-box templates in the Dashboard category.

a Click Dashboard.

b Point to each template and read the descriptions.

7. If you need to create a multipage dashboard style app, which template requires the fewest customizations?

____

Step 6: View the web page and website templates

Finally, you will observe the out-of-the-box templates in the Web Page and Website categories. These two categories of templates are similar; however, Website templates include multiple pages with preconfigured layouts.

- (a) Click Web Page.
- (b) Review the descriptions for up to three Web Page templates. Click
- (c) Website.
- (d) Review the descriptions for the Website templates.

8. What types of information are emphasized in the Web Page and Website templates?

____

- (e) Close Firefox.

In this exercise, you explored the available out-of-the-box templates and identified the appropriate template for a given scenario. In the next exercise, you will create a new experience with a custom template and adjust the template to align with a design.

Choosing an out-of-the-box template category

In the previous activity, you discovered the many available out-of-the-box templates. While you can review the specific capabilities for each template individually, it is helpful to narrow down the selection process by choosing a template category. To choose the appropriate template category, consider the requirements of the planned web app. The possible out-of-the-box template categories are presented in the following list:

- Blank
- Web AppBuilder classic
- Map centric
- Dashboard
- Web page
- Website

You will now identify the template category that addresses a given set of app requirements.

Set of app requirements	Template category
Feature images and text in a multipage app.	
Tailor the layout to meet specific functionality and branding requirements.	
Rebuild a mapcentric ArcGIS web app that includes widgets with ArcGIS Experience Builder.	
Visualize live data using charts, tables, and text.	
Highlight a web scene and 3D functionality.	

Introduction to the interface

Before applying the design principles that you have discovered in this lesson, you will be introduced to the interface of an ArcGIS Experience Builder experience.

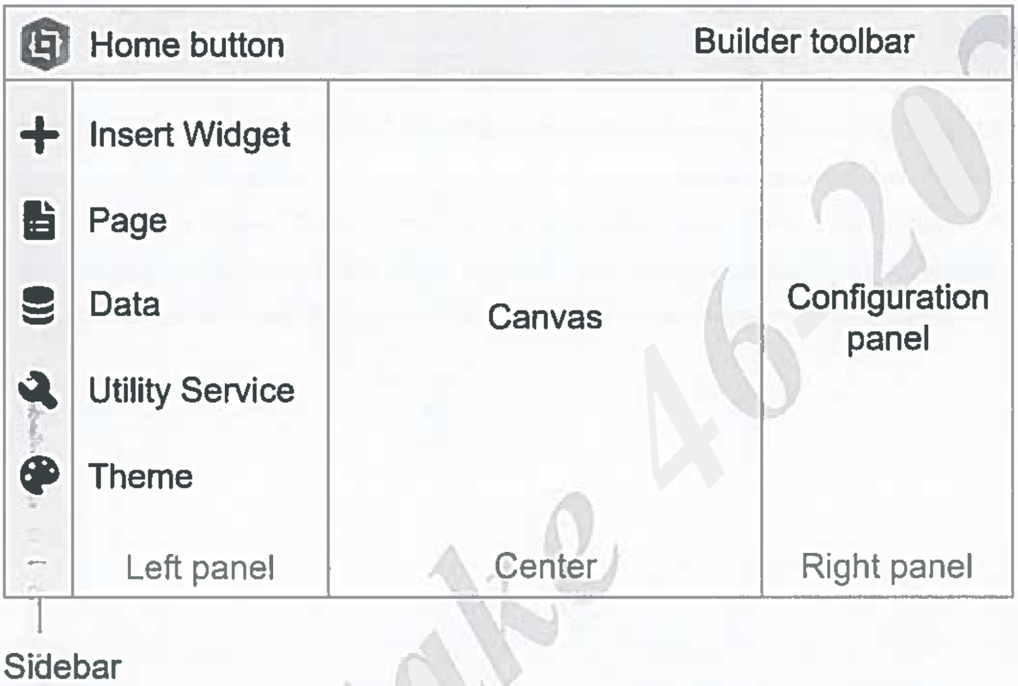


Figure 2.2. In ArcGIS Experience Builder, there are several key interface elements that are used when working with an experience.

Interface sections

An experience has four key sections that are used to create an app based on a design:

Introduction to the interface (continued)

- **Left panel:** The left panel in the experience is activated by the Sidebar. From the Sidebar, the Insert Widget panel, Page panel, Data panel, Utility Service panel, or Theme panel can be opened.
- **Center:** Located in the center of the experience, the Canvas is used to build the layout of your ArcGIS Experience Builder app.
- **Right panel:** The right panel in the experience is referred to as the Configuration panel. The Configuration panel is contextual, meaning that the options on this panel change based on the element that is selected on the Canvas.
- **Toolbar:** The upper portion of the experience contains the Builder toolbar, which provides options for saving, previewing, and publishing your app. To return to the ArcGIS Experience Builder landing page, you can click the Home button, which is located near the top-left corner of the toolbar.

Exercise 2B

25 minutes

Apply design principles to an app

In this exercise, imagine that you are a web designer who needs to create a single-page web app for water resource planners that displays a map of global drought risk. You already created a design that considers the purpose and target audience for the app and addresses the key design considerations. The following graphic presents the key design elements in the web app design.

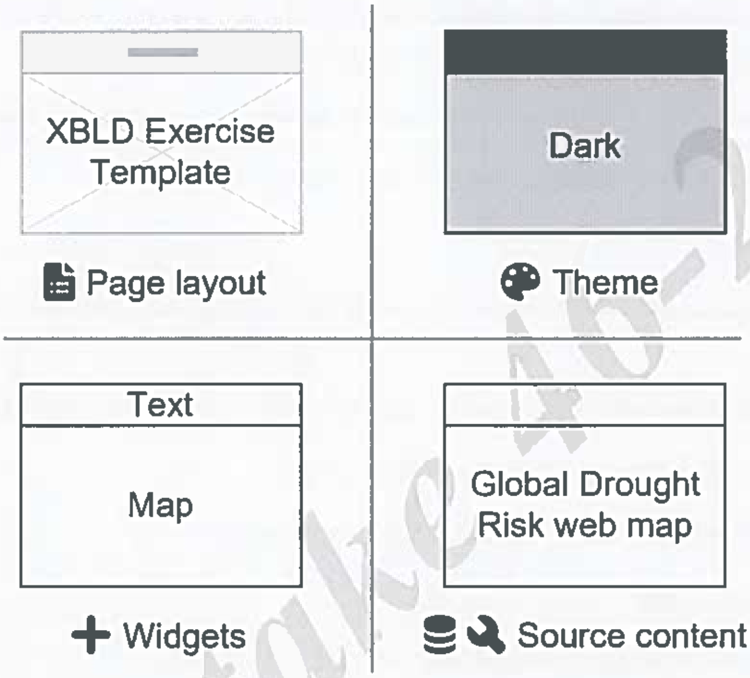


Figure 2.3. This graphic shows the template-defined page layout (XBLD Exercise Template), theme (Dark), widgets (Map and Text), and source content (Global Drought Risk web map) that will be included in the web app.

You will use this design information to configure a basic web app. In this

exercise, you will perform the following tasks:

- Use a template to define the page layout.
- Set the theme for the app framework.
- Adjust the widgets on the page.
- View the source content for a widget.

ArcGIS Experience Builder Help: *Choose the app theme*

Step 1: Open ArcGIS Experience Builder

First, you will use a template to create a single-page web app. As part of the design phase, the following question was considered.

- Which template will be used to create the app?

In this case, the XBLD Exercise Template aligns with the app requirements with some minor customizations. This custom template was created by another member in your organization.

a. Open Firefox, and then on the ArcGIS Experience Builder landing page, click Sign In. For the

b. organization's URL, type **trainingservices**, and then click Continue.

c. Click Your Course Account.

d. Type the organizational account username and password provided by your instructor, and then click Sign In.

You will now create a new experience using a custom template that is shared with ArcGIS Online.

e. Near the top-right corner of the page, click Create New. Under

f. Select A Template To Start, click the ArcGIS Online tab. In the Search

g. field, type **XBLD** and press Enter.

h. Locate the XBLD Exercise Template, and then click Create. If

i. prompted, in the Getting Started window, click Skip.

The new experience with a preconfigured layout opens. This experience is stored in your ArcGIS Online content and is only shared with you. You will now rename the experience to align with the purpose of the web app.

j. In the top-left corner, next to the Home button, click XBLD Exercise Template 1 to activate the name.

k. Delete the current title text, type **Global Drought Risk <your initials>**, and press Enter.

So that you retain a copy of the original preconfigured layout that was included in the template, you will duplicate the current page.

l. From the Sidebar, click the Page button.

Hint: The Sidebar is located on the left side of the experience.

m In the Page panel, point to Page Template, click the More button, and choose Duplicate. Point to

n Page Template 2, click the More button, and choose Rename.

o Rename the page to **Applying Design Principles** and press Enter.

You will now set this new page as the home page, which will make this page the active page when the web app is published. Currently, this app does not meet the configuration requirements for a multipage web app, so the published app will only have one page. You will learn more about configuring a multipage web app later in this course.

p For the Applying Design Principles page, click the Make Homepage button.

In the subsequent steps, you will configure the layout for the single-page web app on the Applying Design Principles page.

Step 2: Set the theme for the app

You will now set the theme for the web app. As part of the design phase, the following questions were considered.

- What are the branding requirements for the app?
- What is the key information in the app?

In this case, there are no branding requirements for the app; therefore, you will choose a theme that best represents the key information in the app.

a From the Sidebar, click the Theme button.

b In the Theme panel, click each theme and view the changes to the Canvas.

Hint: The Canvas is the center area of the experience that shows the layout of the current page in the web app.

You will notice that the color scheme and font change when each theme is applied. The Theme panel sets the default color scheme and font for all pages in the app framework.

An administrator of the ArcGIS Online organization needs to configure the Organization Shared theme. If an organization theme is not configured, the option will show the default theme.

c Click the Dark theme.

You are choosing the Dark theme because it emphasizes the map using a color scheme that aligns with the topic of water resources. Next, you will customize the theme colors.

d At the bottom of the Theme panel, click Customize.

1. What is the primary color for the Dark theme?

____

Hint: Click the icon above Primary to view the color settings. Then, click outside the window to close the color settings.

e Click Advanced Color Setting.

You will notice that the selected theme has a specific color set. In the color set, there are eight color variables: primary, secondary, light, dark, success, info, warning, and danger. Each color variable has nine grayscales that consist of the theme colors for your app to apply to pages and widgets.

f Click the red color variable, as indicated in the following graphic, and choose a purple color.

You will notice that the nine grayscales for the color variable update automatically.

g Change the theme color set by clicking the right arrow, as indicated in the following graphic, and choosing a new color set.

h Click Reset.

The theme reverts back to the default Dark theme.

i Click Back.

You will now customize the theme font.

j For Theme Font, choose Arial.

On the Canvas, you will notice that the font for Page Name changes to Arial. You will change the color of the title font in the next step.

Step 3: Adjust the widgets on the page

Next, you will adjust the widgets on the page and view the source content for a widget. As part of the design phase, the following questions were considered.

- What functionality does the app need to provide?
- Where will the source content come from?

In this case, the web app needs to display an interactive web map from ArcGIS Online and a title.

a From the Sidebar, click the Insert button.

Widgets are added to a page from the Insert Widget panel. The page currently contains two widgets: a Map widget displays a web map, and a Text widget shows the page name.

First, you will adjust the alignment of the Map widget.

b On the Canvas, click the map to select the Map widget. Click the

c Align button and choose Full Width.

The map expands to fill the full width of the page.

To the right of the Canvas, the configuration panel displays the options for Map 2, which is the name of the map widget that you selected. The configuration panel is contextual, meaning that the options on this panel change based on the element that is selected on the Canvas. Later in the course, you will learn how to rename the widgets to meaningful names.

2. What is the name of the web map that is powering the Map 2 widget?

____

Hint: This information is displayed in the Map 2 configuration panel under the Source heading. From

d the Sidebar, click the Data button.

You will notice that the source web map is also listed in the Data panel. The Data panel shows all the source data in the app framework that can be used to power the widgets.

Next, you will adjust the alignment of the Text widget.

e On the Canvas, point to the header (blue banner) and click Edit Header. Click Page

f Name to select the Text widget.

g Click the Align button and choose Horizontal Center.

h In the Text widget configuration panel, under Text Format, click the Center button to format the text to use center justification.

i Under Text Format, click the Font Color button and change the color of the text to use the Dark color variable, as shown in the following graphic.

In the configuration panel, you will also notice that the Text widget is not connected to data.

j On the Canvas, click outside of the header to deactivate it.

Changing the font format in the configuration panel only impacts the selected widget. The theme that is applied at the app framework level (Theme panel) is not altered.

Step 4: View the app on various devices

Finally, you will view the layout on various screen sizes to ensure that the app provides a user- friendly experience for the target audience. As part of the design phase, the following question was considered.

- How will users in the target audience access the app?

In this case, the web app will primarily be accessed by planners on a desktop device.

a At the top of the page, ensure that the Edit Your Page For Large Screen Devices button is active, as shown in the following graphic.

When building a new app using Experience Builder, the default view is the large-screen device view. The large-screen view simulates the user experience with a desktop device, which is the primary way that the app will be accessed by planners in this scenario. To ensure that the app is usable on other types of devices, you will also review the app on a medium- and small-screen device.

b. Click the Edit Your Page For Medium Screen Devices button . Click the

c. Edit Your Page For Small Screen Devices button .

In general, the app is usable on the other screen sizes. Later in this course, you will learn how to customize these views to provide a more mobile-friendly experience.

d. Near the top-right corner of the page, click the Save button to save the experience. Click

e. the Home button to return to the landing page.

f. Leave Firefox open for the next exercise.

In this exercise, you used a design to configure a web app.

Lesson review

1. **What approach could an organization use to provide members a standard layout in ArcGIS Experience Builder?**

2. **Which type of ArcGIS Online members can create custom themes for their organization?**

Answers to lesson 2 questions

Exercise 2A: Explore the out-of-the-box templates (page 2-6)

1. When should a blank template be used?

   Use a blank template to create web apps from scratch based on your own design.

2. Which template provides a fully blank layout that minimizes scrolling?

   Blank Fullscreen; the template includes a single page with a full-screen app layout and no organizational elements.

3. Which ArcGIS Experience Builder template best aligns with the existing Philadelphia Marathon Resources Web App?

   Foldable; it includes a header and a controller widget (in the header).

4. For Scenario 1, which template should be used to create the planned web app?

   JewelryBox

5. For Scenario 2, which template should be used to create the planned web app?

   Route

6. For Scenario 3, which template should be used to create the planned web app?

   Data Collector

7. If you need to create a multipage dashboard style app, which template requires the fewest customizations?

   Indicator

8. What types of information are emphasized in the Web Page and Website templates?

   Text and images

Answers to lesson 2 questions (continued)

Choosing an out-of-the-box template category (page 2-12)

Set of app requirements	Template category
Feature images and text in a multipage app.	**Website**
Tailor the layout to meet specific functionality and branding requirements.	**Blank**
Rebuild a mapcentric ArcGIS web app that includes widgets with ArcGIS Experience Builder.	**Web AppBuilder classic**
Visualize live data using charts, tables, and text.	**Dashboard**
Highlight a web scene and 3D functionality.	**Map centric**

Exercise 2B: Apply design principles to an app (page 2-15)

1. What is the primary color for the Dark theme?

   HEX #0c91f0 or R: 12, G: 145, B: 240, A: 100

2. What is the name of the web map that is powering the Map 2 widget?

   Global Drought Risk

3 Adding pages and windows to an app

In this lesson, you will discover how to configure the layout of the pages and windows in a multipage web app.

Topics covered

- Pages
- Windows

Working with pages

A key capability of ArcGIS Experience Builder is the ability to create web apps that contain one or more pages. When configuring a web app, it is important to understand the anatomy of a page and, if applicable, how the end users will navigate between the pages in the app.

Anatomy of a page

Each page in a web app can consist of a header, body, and footer. The header and footer are optional components that can be turned on or off for each page in an app. While the content of the body will vary on each page, the content in the header and footer are consistent across all pages.

When new pages are added to an app, an out-of-the-box template must be selected for the body. Depending on the selected template, the layout of the page will be sized to fill the device screen (full-screen app) or require scrolling to view all the content (scrolling page).

- Fullscreen apps typically make use of a map that fills the device screen. For example, many of the mapcentric templates are available in this layout category.
- Scrolling pages often include a mix of text, images, and other interactive elements that require more screen space. For example, many of the dashboard and web pagetemplates are available in this layout category.

Working with pages (continued)

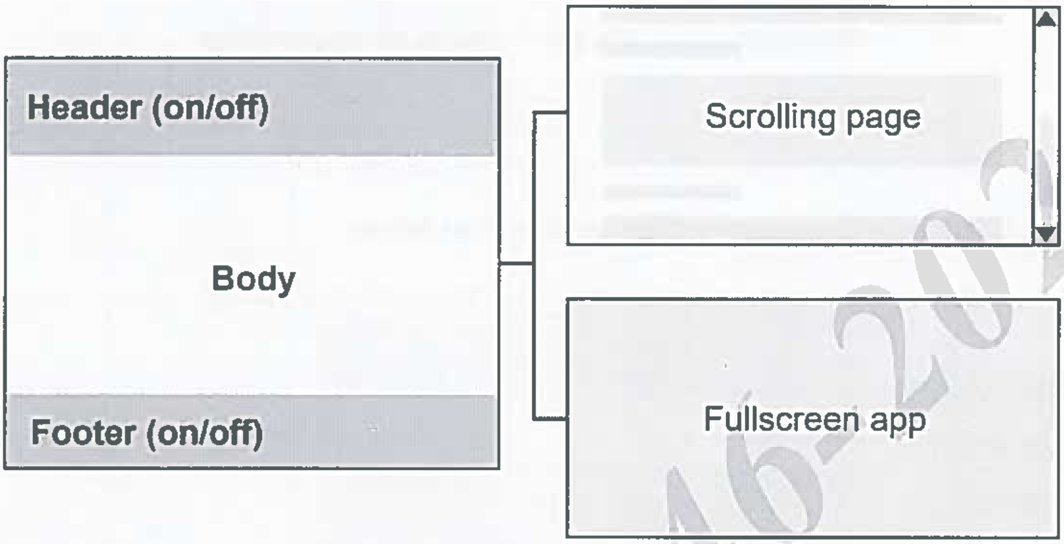


Figure 3.1. The left side of the graphic shows the three components of a page: header, body, and footer. The right side of the graphic shows an example of a scrolling page and full-screen app, which are the two types of layouts for the page body.

Navigating between pages in an app

If you are configuring a multipage app, it is important to determine how the end-user will navigate between the pages in the app. The most common approach for providing this functionality is through the use of a menu in the header.

The menu widget creates a list of page links that enable users to move between the pages in an app. There are two primary ways of displaying the page links in the menu:

- Display a Horizontal or Vertical menu of the page links.
- Show a clickable Icon menu that opens a sidebar window that contains a list of the page links.

Working with pages (continued)

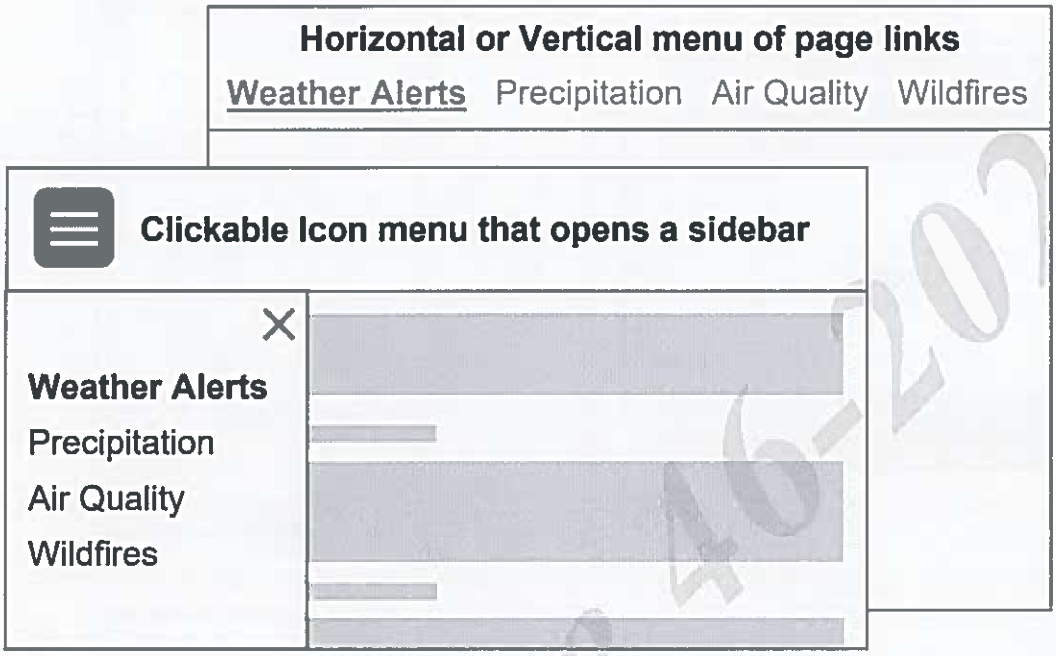


Figure 3.2. This graphic shows the two ways of displaying page links as a menu for a web app. A user can move between the four pages in each web app through the menu.

Exercise 3A

30 minutes

Configure a multipage app

In this exercise, imagine that you are a web app designer who needs to configure a multipage app for the City Historical Society. This new web app should provide the following resources for potential visitors:

- **Activities:** Curated list of historical tours and events
- **History:** Historical timeline for the city
- **Photos:** Photos that show changes to historical sites over time
- **Landmarks:** Bookmarks that enable users to view historical landmarks on a map
- **Virtual Tour:** 3D virtual tour of the city that enables users to fly along routes with historical significance

To ensure that visitors can navigate to all these resources, a link to each page in the web app will be provided by a menu in the header.

In this exercise, you will perform the following tasks:

- Add pages to a multipage app.
- Configure the page components (header, body, and footer).
- Use a menu to move between pages.

Step 1: Create a new experience

First, you will create a new multipage web app for the City Historical Society using the Exhibition template. Located in the Website category, the Exhibition template closely aligns with the purpose and target audience for the planned web app.

a. If necessary, open Firefox and sign in to ArcGIS Experience Builder.

Hint: Sign in to the **trainingservices** organization using your course account. Near

b. the top-right corner of the page, click Create New.

c. On the Default tab, click Website. For

d. Exhibition, click Create.

e. To the right of the Home button, rename the experience to **City Historical Society <your initials>**.

Hint: In the top-left corner, click Untitled Experience # and delete the existing text, and then type the new name and press Enter.

Before you can edit the page layouts in the experience, you must ensure that the layout is unlocked. This option is often enabled when an experience is created from a preconfigured template.

f. On the Builder toolbar, turn off the option for Lock Layout.

Hint: The Builder toolbar is located near the top of the experience.

Because you created the City Historical Society experience from an out-of-the-box template in the Website category, the web app is preconfigured to contain multiple pages. The template includes a header that contains a horizontal menu of page links that enable users to move between the pages in the web app.

Next, you will rename the pages to align with the resource that is being provided.

g. In the Page panel, rename each page based on the information in the following table.

Existing page name	New page name
Page	Overview
Page 2	Activities

Existing page name	New page name
Page 3	Photos
Page 4	Landmarks
Page 5	History

Hint: You can double-click the existing page name to rename the text; alternatively, you can point to the existing page name, click the More button, and choose Rename.

After making these updates, the page links in the header of the web app match the new page names. To improve the flow of information, you will now reorder the pages.

h In the Page panel, select History, and then drag this page above Photos in the list.

The History page is now the third page in the web app.

The order of the page links in the header also updates to show that History is the third page in the web app. From the exercise scenario, you may notice that the Virtual Tour page is missing from the web app. In the next step, you will add this page to the app.

i Near the top-right corner of the page, click the Save button to save the experience.

Step 2: Add a page with a header

When configuring a multipage web app, it is convenient to start with a multipage template. However, you can configure any starting template to include multiple pages by adding pages to the Page panel, turning on the page header in the configuration panel, and configuring a menu in the header. Regardless of the selected starting template, you will likely need to add and remove pages from the experience to meet the app requirements.

You will now discover the process for adding a page with a header to an experience.

a. In the Page panel, click the Add Page button.

You will notice that the available out-of-the-box templates for the new page align with the starting template options for a new app.

b. Under Fullscreen App, select Voyage.

A new page is added to the experience that uses the Voyage template. This new page enables users to tour the city by flying along predetermined routes in a 3D web scene. This template has a Fullscreen App layout, meaning that the 3D scene is configured to fill the device screen.

c. If necessary, in the Page panel, double-click Page to select and edit the current name. Type

d. **Virtual Tour** and press Enter to rename the new page.

Currently, the Virtual Tour page does not contain a header, even though the header is turned on for the other pages in the app. Next, you will turn on the header for the Virtual Tour page.

e. On the right side of the experience, locate the Virtual Tour configuration panel.

As a reminder, the configuration panel is contextual, meaning that it displays the configuration options for the selected element on the active page. If none of the page elements are selected on the Canvas, the configuration options for the active page are shown. In this case, the configuration options for the Virtual Tour page are currently displayed.

f. In the Virtual Tour configuration panel, for Header, turn on the option.

The header is added to the current page. You will notice that the header contains the same information as the Overview page. When the header is edited, the changes are applied to the header throughout the app framework.

Next, you will explore the pages in the web app using the Live View option. This option allows you to preview the interactive functionality in the app while continuing to make changes to the app content.

g. On the Builder toolbar, click Live View. On

h. the Canvas, click Overview.

Instead of providing the option to edit the header, this interaction updated the Canvas to show the Overview page. You will also notice that the configuration panel shows the options that can be modified for this page. Although the full editing options are not available, you can continue to make changes to the configuration options while you are in the Live View mode.

i. On the Builder toolbar, click Live View to return to the full editing mode.

Step 3: Configure the page options

In this step, you will continue to explore the configuration options for pages as you work with the two types of pages in Experience Builder: Fullscreen App and Scrolling Page.

First, you will change the animation for the Landmarks page, which has a Fullscreen App layout.

a. In the Page panel, click Landmarks.

b. In the Landmarks configuration panel, under Animation, click the Preview button.

The page floats in from the top of the page. By default, the exhibition template applies the float in animation for all the pages. While this animation may add to the user experience for some of the pages, this animation may be distracting on the pages that feature maps. You will remove the animation for the Landmarks page.

c. Click Change.

d. In the Animation Settings window, select None.

Next, you will configure the History page, which has a Scrolling Page layout. For pages that require a lot of scrolling, it can be helpful to configure the header to stay at the top of the page. You will apply this configuration option to the History page now.

e In the Page panel, click History.

f In the History configuration panel, under Header, check the box for Stays At The Top Of The Page When Scrolling.

Turning off the footer can also reduce the amount of scrolling. In this case, the footer content is not required on the History page and will be removed.

g For Footer, turn off the option.

h Save the experience.

Hint: Near the top-right corner of the page, click the Save button.

Now that you have configured several page options, you will preview your changes.

i On the Builder toolbar, click the Preview button. On the

j header, click Landmarks.

You will notice that the entrance of the Landmarks page is not animated.

k Scroll to the bottom of the Landmarks page.

You will notice that the footer is included below the map. Footers frequently contain content such as links to other pages, social media links, contact information, and much more. Even though the Landmarks page is configured as a full-screen app, some scrolling is required due to your device resolution and the presence of a footer.

l On the header, click History.

The timeline floats in from the top of the page. In this case, the animation fits with the content that is being presented on the page.

m Scroll to the bottom of the History page.

You will notice that there is not a footer at the bottom of this page.

n Close the current web browser tab.

Step 4: Update the page outline

The Page panel displays all the widgets that are included in the header, body, and footer in the outline section. It is helpful to rename the widgets in the outline to meaningful names so that you understand how these elements are being used in the app.

a In the Page panel, click History to select it, if necessary, and then under Outline, expand the Header section.

You will notice that the Header contains three widgets: Menu, Image, and Button.

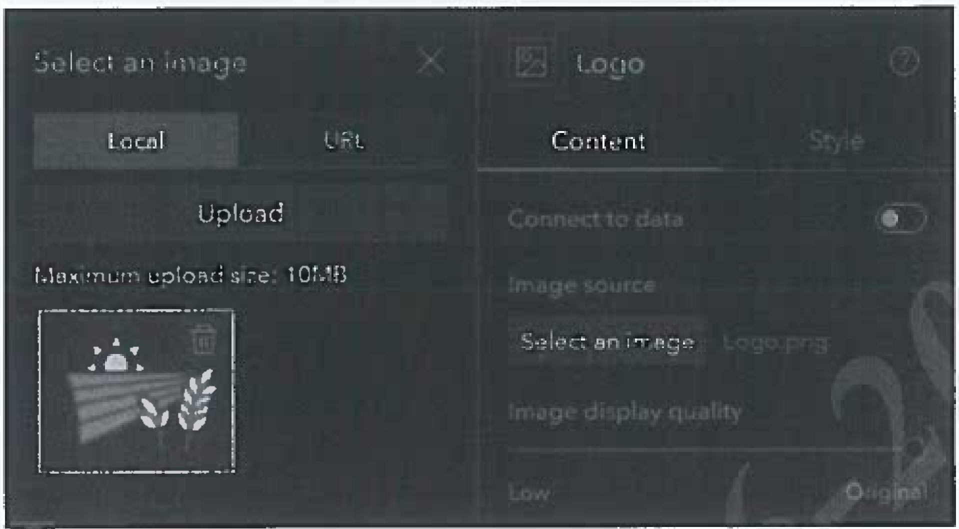
b Rename Image to **Logo**.

In this case, Image will be used to display the logo for the historical society. You will now add the logo to the widget.

c In the Logo configuration panel, under Image Source, click Select An Image. In the

d Select An Image window, on the Local tab, click Upload.

e. Browse to **C:\EsriTraining\XBLD\City Historical Society**, select the Logo.png file, and click Open.



When an image is selected and not connected to data, it is stored within the application structure. Optionally, you can add an image that is connected to data, which is a type of source content. You will learn about connecting widgets to data later in this course.

The image source for the widget now references the Logo.png file, and the Canvas updates to show the logo for the historical society.

f. In the Page panel, rename Menu to **Header Menu**.

In the next step, you will configure the Menu widget in the header.

Step 5: Configure the menu

A menu is a key element in a multipage app. Menus provide a list of page links that allow users to move between the pages in the web app. In this step, you will explore the available configuration options for the Menu widget.

First, you will set icons for the page links in the menu.

a. In the Header Menu configuration panel, under Appearance, turn on the option for Show Icon. In the

b. Page panel, select the Landmarks page.

c. Point to Landmarks, click the More button, and choose Set Icon.

d. Under General, select the Pin icon.

Hint: Point to the icon to see its name.

- **e** Under Header, click Header Menu to select it on the Canvas.
- **f** On the Canvas, expand the Menu widget so that you can see all the page links.

Next, you will change the style of the menu. Currently, the menu is configured with the underline style, meaning that the active page in the web app is underlined in the menu.

- **g** In the Header Menu configuration panel, under Appearance, for Style, select Default.

You will notice that the underline that indicates the current page disappears. By default, the active page is not identified in the menu.

- **h** For Style, select Pills.

You will notice that the active page is now encased in an oval shape in the menu. In this exercise, you will select the Pills style for the menu. Next, you will modify the advanced menu properties.

- **i** For Advanced, turn on the option.

The Advanced section appears. This section contains the following three tabs:

- **Default:** This tab displays the properties for the inactive pages.
- **Selected:** This tab displays the properties for the active page.
- **Hover:** This tab displays the properties for each page when a user points to that link.

- **j** On the Default tab, under Text, click the text color to change it.
- **k** In the color picker, under Theme Colors, point to each color in the first row.

You will notice that the name of each color variable is displayed. As a reminder, the first row in the theme color set displays the eight standard color variables. The next nine rows display the grayscales for each color variable. In the course exercises, you will be directed to choose a color based on its name in the color picker. In this case, you will apply the Secondary color.

l Under Theme Colors, in the second column, click the Secondary color.

You will notice that the text for the inactive pages (Overview, Activities, History, Photos, and Virtual Tour) changes to a light gray color.

m Under Advanced, click the Selected tab.

n On the Selected tab, for Background, click the fill color to change it.

o In the color picker, under Theme Colors, in the first column, in the fourth row, click the Primary-300 color.

You will notice that the color of the pill (oval shape) changes to a medium gray color.

So far, you have been using the horizontal menu in the header, which displays the page links as a horizontal list. You will now explore the icon menu. The icon menu displays a button that opens a sidebar with the list of page links.

p In the top portion of the Header Menu configuration panel, for Type, click the drop-down list and choose Icon.

The menu is now displayed as a button, rather than a series of page links.

q On the Builder toolbar, click Live View.

r On the Canvas, near the upper-left corner, next to the logo, click the Menu button.

A sidebar window with the page links appears.

s In the sidebar window, click Overview.

You will notice that Overview becomes the active page on the Canvas.

t On the Builder toolbar, turn off Live View, and then save the experience.

u. Click the Home button.

v. Leave Firefox open for the next exercise.

In this exercise, you configured the pages in a multipage app.

Working with windows

Like pages, windows are used to display content in a web app. You can add windows to a web app as splash screens and set links. When a new splash screen or set link window is added to an app, an out-of-the-box template must be selected. When the window is created, widgets can be added and removed from the window to meet the app requirements.

Splash screen windows

A splash screen appears when a user opens the app. A splash screen can contain user agreements, notifications, or other types of information that all users accessing the app need to know. The splash screen setting must be configured for a window for it to display in this manner.

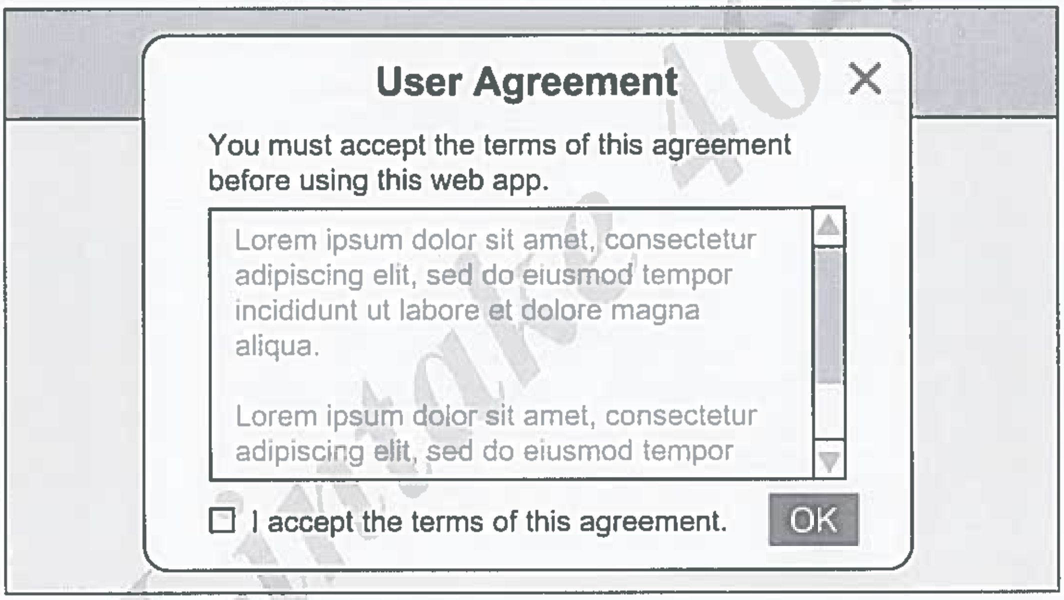


Figure 3.3. This example shows a splash screen that contains a user agreement.

Windows that open with a button

You can create custom buttons that open a window using a widget. Custom buttons are often used to present helpful or contextual information. When a user clicks the button, the window appears on the screen.

Custom buttons can also be configured to open pages, section views, and external windows in an app.

Working with windows (continued)

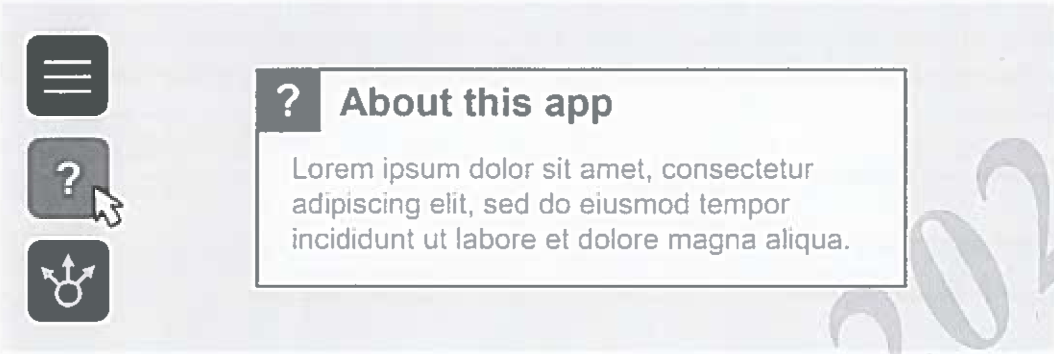


Figure 3.4. In this example, the About This App window is opened by clicking a custom button.

Exercise 3B

25 minutes

Add windows to an app

In this exercise, imagine that you are a web app designer who needs to configure a multipage app for the City Historical Society. You will build on the web app that you created in the previous exercise by adding a splash screen that welcomes visitors and then creating a custom button that contains information about the historical society.

In this exercise, you will perform the following tasks:

- Create a welcome splash screen.
- Configure an About Us button.

Step 1: Add a splash screen

In this step, you will add a splash screen to the experience by adding a window and then setting it as a splash screen. First, you will add a window to the experience.

a. If necessary, open Firefox and sign in to ArcGIS Experience Builder. Click

b. City Historical Society <your initials> to open the experience.

> ⚠ The instructions in this box are only necessary if you did not already complete the previous workflow to create the City Historical Society experience. If you are unsure, ask your instructor for assistance.
>
> 1. Click Create New, and then click Website.
> 2. For Exhibition, click Create.
> 3. Double-click Untitled Experience # and delete the existing text, and then type **City Historical Society <your initials>** and press Enter.
> 4. Click the Save button.

c. From the Sidebar, click the Page button. In the

d. Page panel, click the Window tab.

You will notice that the Exhibition template already contains a window. Starting templates can include preset pages and windows.

e. Click the Add Window button.

In this case, you want to use the splash screen to welcome users to the website. You will choose the Simple template for the splash screen window.

f. In the Add Window window, select Simple.

g. If necessary, in the Page panel, double-click Window 2 to select and edit the current name. Type **Splash**

h. **Screen** and press Enter to rename the new window.

i. Point to Splash Screen and click the Set As Splash button.

Because the Splash Screen window is selected in the Page panel, it is displayed on the Canvas.

You will now customize the look of the window by adding a mask and rounding the edges.

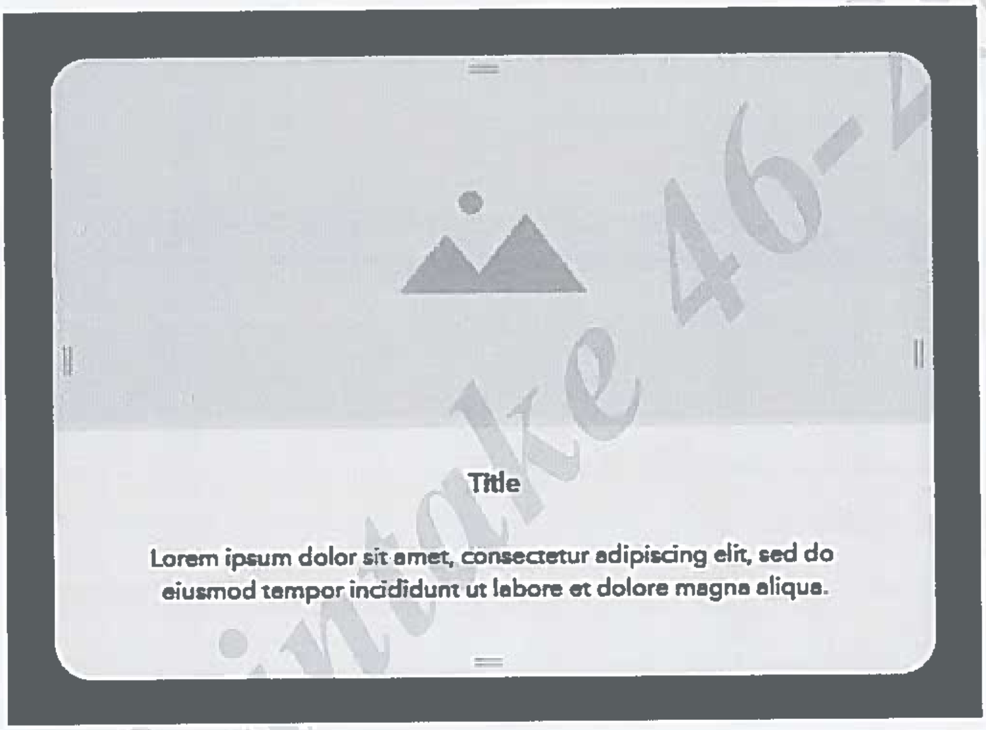
j In the Splash Screen configuration panel, for Mask Color, change the color to #484848, Primary.

Hint: In the first row of the color picker, click the color in the first column.

You will notice that applying a mask changes the color of the interface around the window. A mask can be used to draw attention to the window content.

k For Border, click the Quick Style button and choose Custom. For

l Border Radius, type **20** px and press Enter.

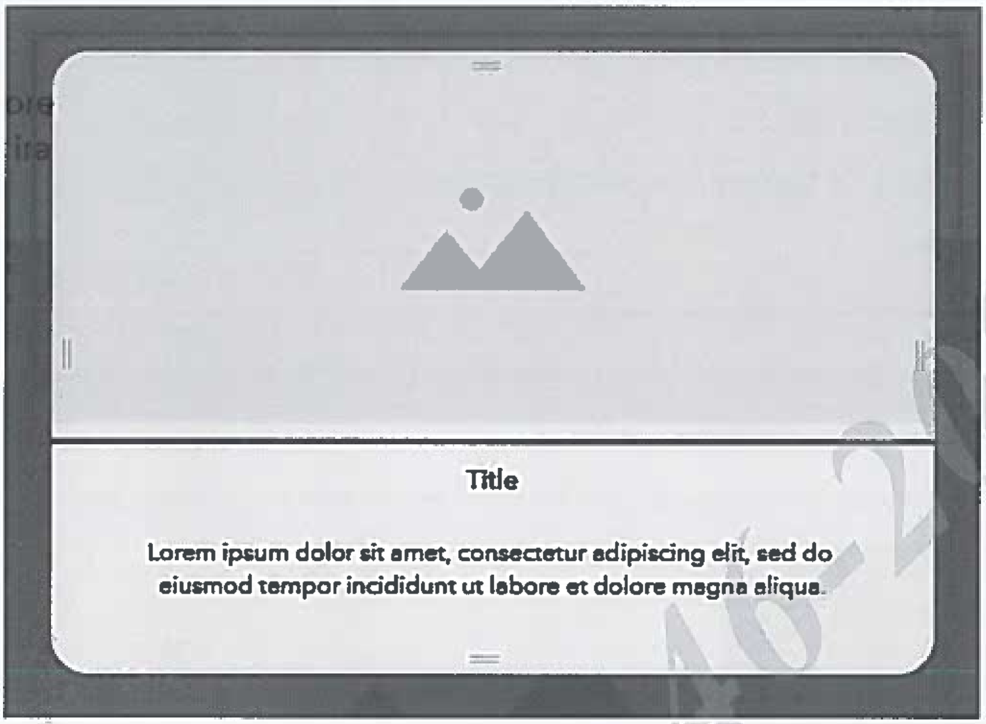


The edges of the window become rounded. Now that you have customized the look of the window, you will populate the splash screen content.

Step 2: Populate the content in a window

In this step, you will populate the content in the splash screen window. First, you will configure the existing image and text widgets in the window.

a On the Canvas, click the image placeholder, as indicated in the following graphic.



b In the Image widget configuration panel, under Image Source, click Select An Image. In the

c Select An Image window, on the Local tab, click Upload.

d Browse to **C:\EsriTraining\XBLD\City Historical Society**, select the Welcome.png file, and click Open.

The image is added to the splash screen window.

e On the Canvas, click Title to select it and press Delete.

You will notice that the existing text displays lorem ipsum. Lorem ipsum is placeholder text that is commonly used to illustrate the visual form of a document without relying on meaningful content. You will now replace the placeholder text with meaningful content.

f Double-click the existing text in the window.

g Type **Use this website to explore the many historical sites and activities in City.**

h If necessary, resize the text box so that the text fits on one line. On the

i Canvas, click outside of the window to apply the new text.

Next, you will configure the window options to provide ways for a user to close the window.

j. In the Splash Screen configuration panel, under Close Options, verify that the box for Click Outside To Close Window is checked.

k. For Interaction, turn on the option.

l. Under Style, select Button In The Same Line (middle option). For

m. Button Text, type **Close**.

   Hint: Click outside of the field to apply the change.

You will also provide users the option to not show the splash screen every time that the app is accessed.

n. Select the "Don't Show This Again" Option.

o. For Checkbox Text, replace the existing text with **Do not show this welcome window again.**

p. On the Canvas, adjust the layout of the widgets in the window so that the information does not cut off or overlap.

You finished configuring the splash screen window, so now you will close the window.

q. In the Page panel, click the Page tab.

Step 3: Create a custom button

In this step, you will add a custom About Us button to the experience by adding a window, populating the content, and then connecting the window to a Button widget. First, you will add a window to the Page panel.

a. In the Page panel, click the Window tab.

b. Click the Add Window button, and then select Sidebar. Type

c. **About Us** to name the new window.

Next, you will populate the content in the About Us window.

d. On the Canvas, scroll to the right, if necessary, select the image placeholder, and then add the Welcome graphic.

Hint: In the configuration panel, click Select An Image, and then click the image that you added in the previous step.

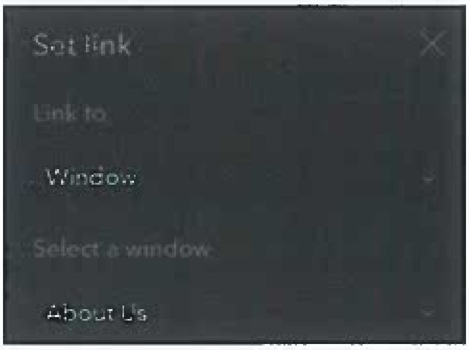
- **e** Close the Select An Image window.
- **f** On the Canvas, double-click Title, and then type **About Us**.

  Hint: Click outside of the window to apply the change to the text.

- **g** Expand the width of the window by dragging the left side of the window to the left.

For now, you will leave the lorem ipsum text as a placeholder in the window. Next, you will configure a Button widget to connect to the new window.

- **h** In the Page panel, click the Page tab to close the About Us window. If
- **i** necessary, under Outline, expand the Header section.
- **j** Under Header, rename Button to **About Us Button**.
- **k** In the About Us Button configuration panel, click Set Link.
- **l** In the Set Link window, perform the following steps to set the link:
  - For Link To, choose Window, if necessary.
  - For Select A Window, choose About Us.



(m) Click OK to close the Set Link window.

The About Us window is now connected to the About Us Button widget.

Step 4: Add an interactive effect to a button

In this step, you will make the About Us button interactive by adding a hover effect.

(a) In the About Us Button configuration panel, for Text, type **About Us**. For

(b) Advanced, verify that the option is turned on.

In the Advanced section, you will define the appearance of the button when a user is not interacting with the button (Default tab).

(c) Under Advanced, on the Default tab, for Fill, change the color to #C5C5C5, Secondary.

Hint: In the color picker, Secondary is one of the color variables that is listed in the first row, under Theme Colors.

Next, you will define the appearance of the button when a user is hovering a cursor over the button (Hover tab).

(d) Under Advanced, click the Hover tab, and then perform the following steps:

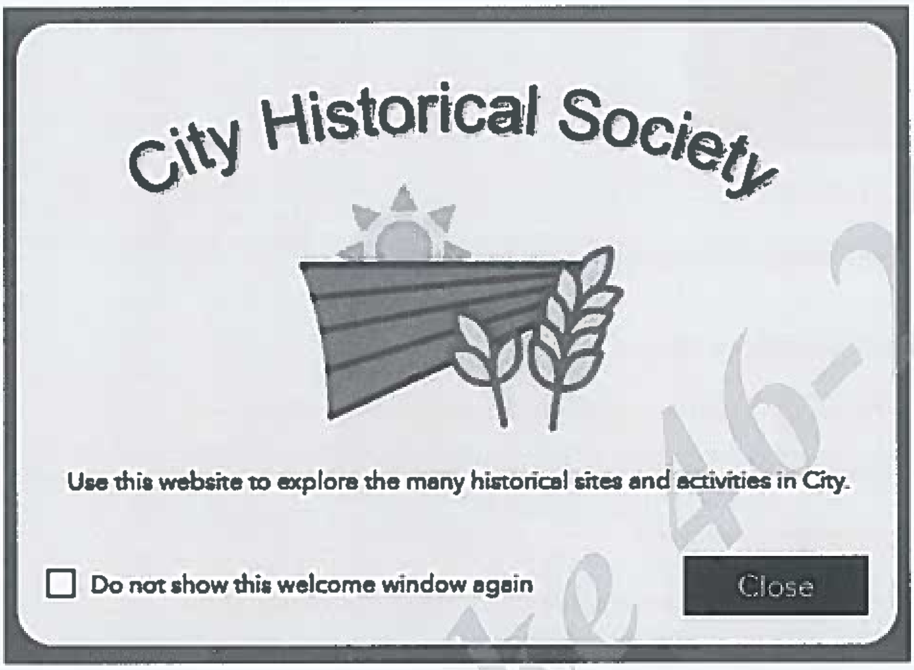
- For Text, click the Bold button, and then change the color of the text to #F0F0F0, Light.
- For Fill, change the color to #089BDC, Info (blue color).

You modified the appearance of the About Us button by adding an interactive effect to the button.

Step 5: Preview the windows

You will now preview the app to test the functionality of the windows in the web app.

a. On the Builder toolbar, click the Save button, and then click the Preview button.



The splash screen appears.

b. Check the box for Do Not Show This Welcome Window Again, and then click Close. In the

c. header, point to About Us.

The button fill color and text change when you hover your cursor over the button.

d. Click About Us.

The sidebar window opens and displays the information about the City Historical Society.

- **e** Close the current web browser tab.
- **f** Click the Home button.
- **g** Leave Firefox open for the next exercise.

In this exercise, you created a splash screen window and an About Us button that opens an informational window.

Linking to an external window

In addition to connecting a custom button to windows in ArcGIS Experience Builder, custom buttons can also link to external windows. The external window can be configured to open in the same window as the app, the top window, or a new window. Examples of external windows include a web address, email, phone number, SMS text message, and FTP resource.

When setting the link for the external window, the appropriate scheme must be specified. The following graphic shows the scheme for each type of external window.

Linking to an external window (continued)

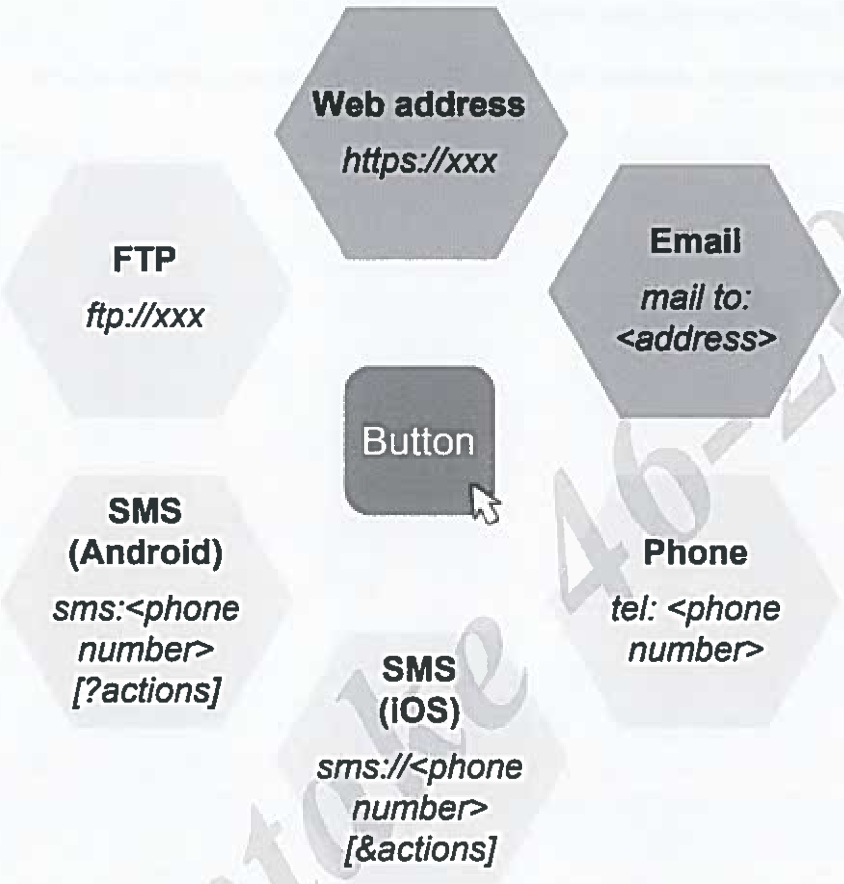


Figure 3.5. This graphic shows the scheme for several types of external windows.

Linking to an external window (continued)

ArcGIS Experience Builder Help: *Button widget*

ArcGIS Experience Builder Help: *Add and connect widgets (includes scheme information)*

Lesson review

1. **Which widget should be included in an app that has multiple pages?**

   a. List

   b. Button

   c. Menu

   d. Widget Controller

2. **Which configuration option needs to be set for a window that will appear when a user first opens the app?**

4 Working with maps

This lesson introduces the Map widget, which is an important widget that is commonly included in web apps. You will discover the process for inserting a Map widget and configuring the widget to connect to a web map.

Topics covered

- Map widget
- Adding data

Overview of the Map widget

Widgets are not only a key design element, but widgets are also the building blocks of an app. This lesson focuses on the Map widget, which is an important building block in ArcGIS Experience Builder. The Map widget is powered by data to display a web map or web scene. On the page, the Map widget can be configured to provide common functionality by turning on standard map tools. Additionally, the Map widget can be connected to other widgets to provide enhanced functionality, which you will learn about later in this course.

The Map widget follows the standard workflow for working with widgets. First, the widget is inserted on a page, and then the widget is configured to provide the required functionality.

As part of the second step in the workflow, the widget is configured. The configuration options for each widget vary based on the function of the widget. For example, source data and tools are two key configuration options for the Map widget. However, there are three possible tabs that organize the configuration options for all widgets:

- **Content:** Defines the source content, tools, behavior, links, and other settings for populating the widget
- **Style:** Customizes the appearance of the widget, including the size, position, animation, background, border, and box shadow
- **Action:** Determines how the widget will interact with the user and other widgets in the app

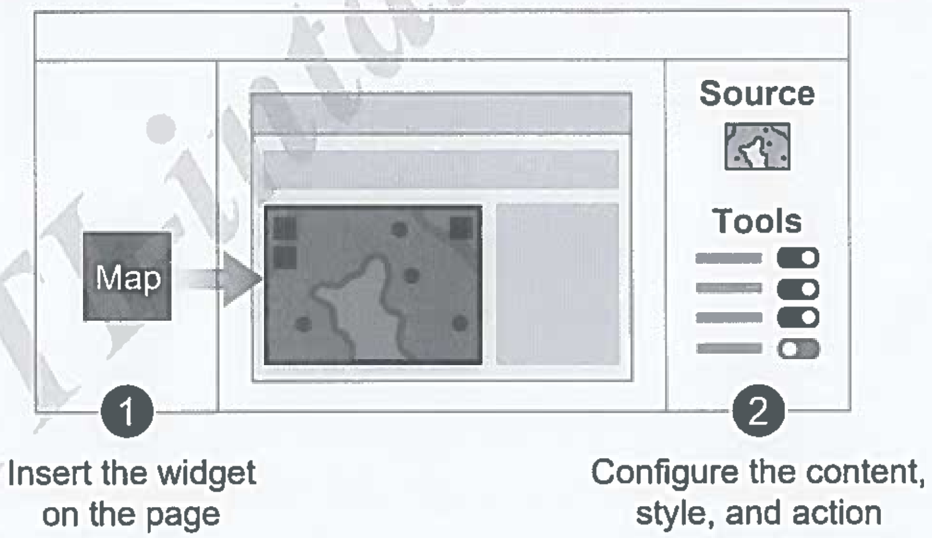


Figure 4.1. This graphic shows the workflow for working with the Map widget. The data source and tools are two key content configuration options for the Map widget.

Explore the map tools

When working with a Map widget, it is important to understand the functionality that the map needs to provide in the app. Configuring the Map widget to include standard tools, which display in the map frame as buttons, is one way to provide this functionality.

In this activity, you will explore the tool configuration options for the Map widget to discover the available functionality.

Instructions

a. If necessary, open Firefox and sign in to ArcGIS Experience Builder. Open

b. your Global Drought Risk experience.

> If you did not already create the Global Drought Risk experience, click Create New. Then, on the ArcGIS Online tab, search for **XBLD**, and then locate XBLD Exercise Template and click Create.

c. On the Canvas, click the map to select it. On

d. the Builder toolbar, click Live View.

e. In the Map configuration panel, under Tools, turn off all the tools.

f. Turn on each tool, and then on the Canvas, explore the tool's functionality.

> You can explore the tool functionality on the Canvas by clicking the button that appears. The Navigation tool may not display a button because you are working with a 2D web map, rather than a 3D web scene.

g. If the tool provides a required function, list the name of the tool in the following table.

h. When you are finished, click the Home button and choose to leave the experience without saving changes.

Required app functionality	Tool
Changing the scale of the map	
Resetting the map to the initial extent	

Explore the map tools (continued)

Required app functionality	Tool
Finding locations of interest	
Viewing a legend for the map	
Changing the reference information on the map	
Determining the distance between two locations on the map	

Adding data to an app

In ArcGIS Experience Builder, the Map widget is powered by data. The process for configuring a Map widget with a data source follows the general two-step workflow for connecting a widget to data. First, the data is added to the app framework through the Data panel. Then, the data is connected to a widget in the configuration panel. At the framework level, the layers and widgets that are connected to each data source can be viewed.

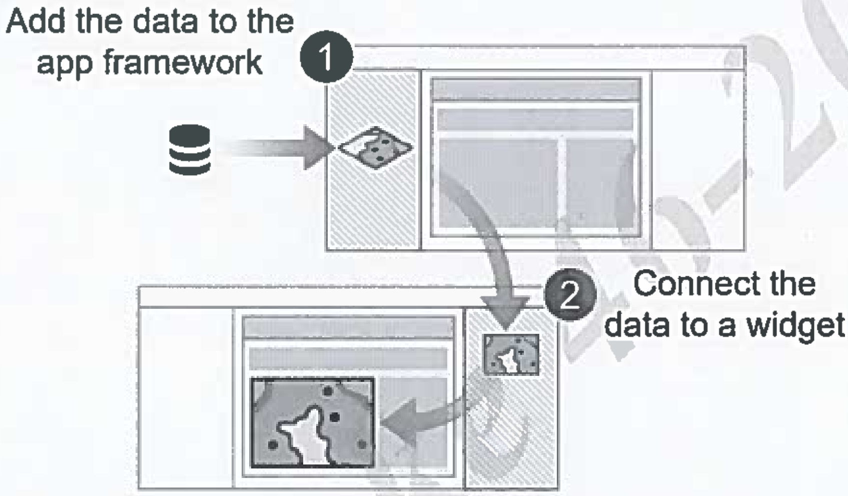


Figure 4.2. This graphic shows the workflow for connecting a Map widget to a web map in ArcGIS Online.

Exercise 4

25 minutes

Build a Map widget with data

In this exercise, imagine that you are a GIS professional for a water management district, and you need to create an app that highlights drought risk in the Rio Grande watershed. This app should highlight a map with tools that enable constituents of the district to explore drought risk.

In this exercise, you will perform the following tasks:

- Insert a widget.
- Configure a widget.

Step 1: Add a Map widget to a blank page

First, you will create an experience called Rio Grande Drought Risk that contains a Blank Fullscreen page.

a. If necessary, open Firefox and sign in to ArcGIS Experience Builder.

   Hint: Sign in to the **trainingservices** organization using your course account. Near

b. the top-right corner of the page, click Create New.

c. On the Default tab, click Blank. For

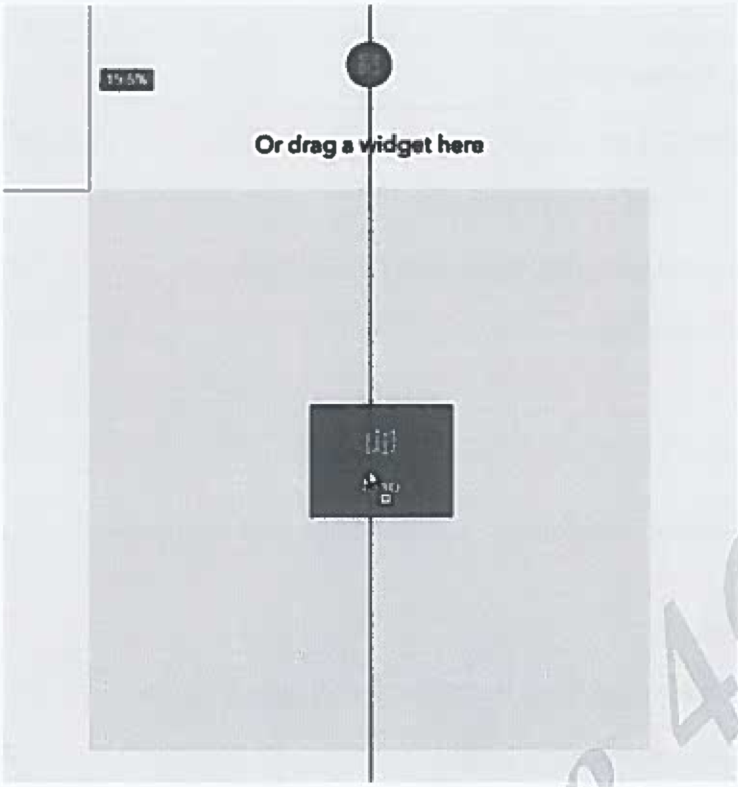
d. Blank Fullscreen, click Create.

e. To the right of the Home button, rename the experience to **Rio Grande Drought Risk <your initials>**.

Next, you will add a Map widget to the page.

f. In the Insert Widget panel, in the Map Centric category, locate the Map widget. Drag the

g. Map widget onto the Canvas, as shown in the following graphic.



h On the Canvas, click the Align button and choose Full Size.

i In the Map configuration panel, at the top of the panel, click Map to edit the name.

j Delete the existing text, type **Drought Risk Map**, and press Enter.

Although the configuration options for each widget vary, the options for every widget are organized into three tabs: Content, Style, and Action (if applicable). For the Map widget, you will start by configuring the options on the Content tab.

Step 2: Connect the map to data

In this step, you will connect the Map widget to a web map in ArcGIS Online. First, you will add the source web map to the app framework. In ArcGIS Experience Builder, data is added to the app framework through the Data panel.

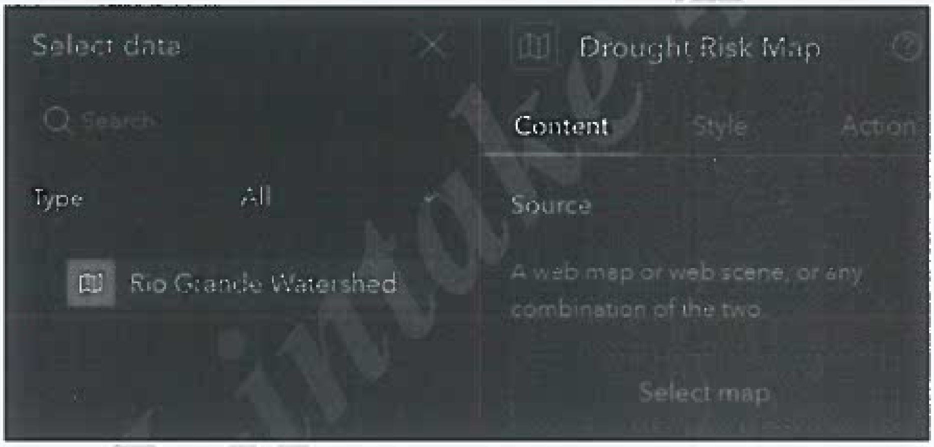
a From the Sidebar, click the Data button to open the Data panel.

Before you can select a map for the widget, you must add the data to the app framework in the Data panel. There are two approaches for adding the data to the Data panel: directly or indirectly. Previously, your instructor demonstrated the process of adding data directly to the panel. You will use the indirect approach in this exercise.

- **b** In the Drought Risk Map configuration panel, under Source, click Select Map. At the
- **c** bottom of the Select Data window, click Add New Data.

When adding new data for a map, you have the option to add a web map or a web scene. In this case, you will add a web map that was created by a member who is not part of your ArcGIS Online organization.

- **d** In the Add Data window, click the ArcGIS Online tab.
- **e** In the Search field, type **Rio Grande Watershed** and press Enter.
- **f** Click the Rio Grande Watershed web map that is owned by EsriTrainingSvc to select it. Near the
- **g** bottom-right corner of the Add Data window, click Done.
- **h** In the Select Data window, notice the Rio Grande Watershed web map that has been added.



The web map is added to the Select Data window as well as the Data panel. Although the web map is now part of the app framework in the Data panel, you will see that the Drought Risk Map widget is still displaying the placeholder map on the Canvas. You will now connect the web map to the Drought Risk Map widget.

- **i** In the Select Data window, click Rio Grande Watershed to select it.

The web map is now listed under Source in the Drought Risk Map configuration panel. You will also see that the Canvas now displays the content of the web map.

Step 3: Modify the map view

In this step, you will change the map extent and display in ArcGIS Experience Builder without modifying the source data in ArcGIS Online. First, you will configure the initial view of the map.

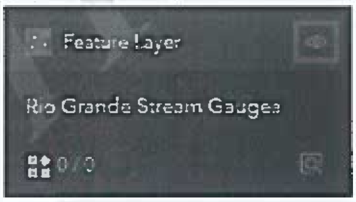
- **a** In the Drought Risk Map configuration panel, under Initial View, select Custom. Under
- **b** Custom, click Modify.

You could pan and zoom in the Modify Initial View window to change the default extent of the map; however, for the purpose of this exercise, you will accept the current extent.

- **c** In the Modify Initial View window, click Cancel.

You will also modify the visible layers in the web map that are connected to the Drought Risk Map widget.

- **d** In the Data panel, click Rio Grande Watershed to view the layers in the map.
- **e** On the Layers tab, for the Rio Grande Stream Gauges layer, click the Hide button, as indicated in the following graphic.



The stream gauges are no longer displayed in the map.

Step 4: Add tools to the map

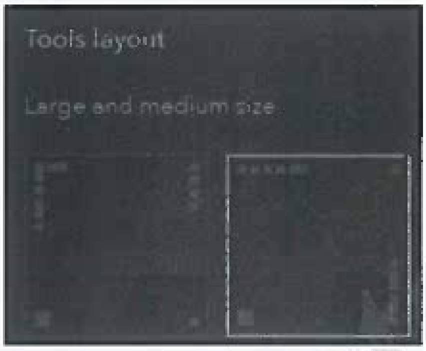
In this step, you will configure the tools that will display in the map and provide the required app functionality.

a. In the Drought Risk Map configuration panel, under Tools, turn on the following tools:

- Layers
- Basemap
- Measure

You will also reposition the tools on the map frame.

b. Under Tools Layout, in the Large And Medium Size section, click the second option, as shown in the following graphic.



On the Canvas, the locations of the tools change to align with the selected wireframe.

Step 5: Configure the map style

In this step, you will style the map by configuring the size, position, border, and box shadow.

a. In the Drought Risk Map configuration panel, click the Style tab. Under

b. Size & Position, for Width, choose Custom.

c. For Width, type **95%** and press Enter.

d. Repeat this process to change the Height to 95%.

e. Above Size & Position, click the Horizontal Center button.

f Click the More Options button for the alignment options, as indicated in the following graphic.

g Choose Vertical Center.

h For Border, click the Quick Style button and choose a border of your preference. For Box

i Shadow, click the Quick Style button and choose Small.

The map frame is now configured to fill 95 percent of the height and width of the screen and is styled with a border and box shadow.

j Click the Save button to save the experience.

k Close Firefox.

In this exercise, you inserted a Map widget to an app page and configured the map content and style. Later in this course, you will explore the Actions tab.

Lesson review

1. **In which two ways can data be added to the app framework?**

2. **When configuring a Map widget, which tool can be turned on for the map?**

   a. Draw

   b. Coordinate Conversion

   c. Search

   d. Suitability Modeler

Answers to lesson 4 questions

Explore the map tools (page 4-3)

Required app functionality	Tool
Changing the scale of the map	**Zoom**
Resetting the map to the initial extent	**Home**
Finding locations of interest	**Search**
Viewing a legend for the map	**Layers**
Changing the reference information on the map	**Basemap**
Determining the distance between two locations on the map	**Measure**

Appendix A

Answers to lesson review questions

Answers to lesson 1 review questions

1. Which ArcGIS Experience Builder version enables you to build custom widgets?

   c. ArcGIS Experience Builder developer edition

2. What capabilities distinguish ArcGIS Experience Builder from other app builders?

   Possible answers include the following capabilities:

   - **Integrating data besides mapcentric content**
   - **Creating multipage web apps**
   - **Designing the layout of the content**

Answers to lesson 2 review questions

1. What approach could an organization use to provide members a standard layout in ArcGIS Experience Builder?

   Create a template and then share the template with the organization.

2. Which type of ArcGIS Online members can create custom themes for their organization?

   Administrators

Answers to lesson 3 review questions

1. Which widget should be included in an app that has multiple pages?

   c. Menu

2. Which configuration option needs to be set for a window that will appear when a user first opens the app?

   Splash screen

Appendix A

Answers to lesson review questions (continued)

Answers to lesson 4 review questions

1. In which two ways can data be added to the app framework?

   Data can be added to the app framework through the Data panel and the widget configuration panel.

2. When configuring a Map widget, which tool can be turned on for the map?

   c. Search

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